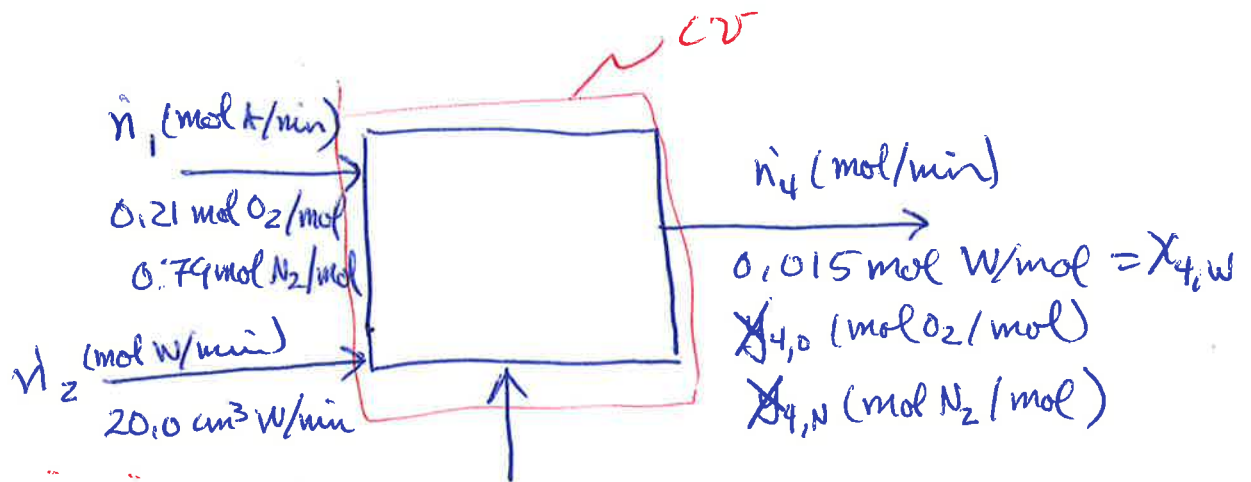


(2)



UNKS: $\dot{n}_1, \dot{n}_2, \dot{n}_3, \dot{n}_4, x_{4,O}, x_{4,N}$ (6) $\dot{n}_3$ (mol/min)

3 balances, 2 constraints, 1 specified stream ($\dot{n}_2$) (6) $\dot{n}_3 = \frac{1}{5} \dot{n}_1$ (constraint)

$$\dot{n}_2 = 20.0 \frac{\text{cm}^3 \text{H}_2\text{O}}{\text{min}} \cdot 1.00 \frac{\text{g H}_2\text{O}}{\text{cm}^3} \cdot \frac{1}{18.015 \text{ g H}_2\text{O/mol H}_2\text{O}} = \boxed{1.11 \frac{\text{mol H}_2\text{O}}{\text{min}} = \dot{n}_2}$$

H₂O Balance $\dot{n}_2 = x_{4,H_2O} \dot{n}_4$

$$\dot{n}_4 = \frac{\dot{n}_2}{x_{4,H_2O}} = \frac{1.11 \text{ mol H}_2\text{O/min}}{0.015 \text{ mol H}_2\text{O/mol tot}}$$

$$\boxed{\dot{n}_4 = 74.1 \frac{\text{mol}}{\text{min}}}$$

Total Balance

$$\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4 \quad \text{use } \dot{n}_3 = \frac{\dot{n}_1}{5}$$

$$\frac{6}{5} \dot{n}_1 + \dot{n}_2 = \dot{n}_4$$

$$\dot{n}_1 = \frac{5}{6} (\dot{n}_4 - \dot{n}_2) = \frac{5}{6} (74.1 - 1.11) \frac{\text{mol}}{\text{min}}$$

$$\boxed{\dot{n}_1 = 60.8 \text{ mol/min}}$$

$$\boxed{\dot{n}_3 = \frac{1}{5} \dot{n}_1 = 12.2 \text{ mol/min}}$$

N_2 Balance

$$x_{N,N} \dot{n}_1 = x_{4,N} \dot{n}_4$$

$$(0.79)(60.8) = x_{4,N} \left(74.1 \frac{\text{mol}}{\text{min}} \right)$$

$$\boxed{x_{4,N} = 0.648}$$

$$x_{4,w} + x_{4,N} + x_{4,o} = 1$$

$$x_{4,o} = 1 - x_{4,w} - x_{4,N}$$

$$= 1 - 0.015 - 0.648$$

$$\boxed{x_{4,o} = 0.337}$$